12da Clase 31/10/2025

75.43 Introducción a los Sistemas Distribuidos
95.60 Redes y Aplicaciones Distribuidas
TA048 Redes

Tema: Capa Física Inalámbrica (II)

Desde *7.4 Cellular Internet Access* hasta el final del Capítulo 7; *Computer Networking: A Top-Down Approach. James Kurose and Keith Ross. Publisher: Pearson, Edition: 8th, 2021.*

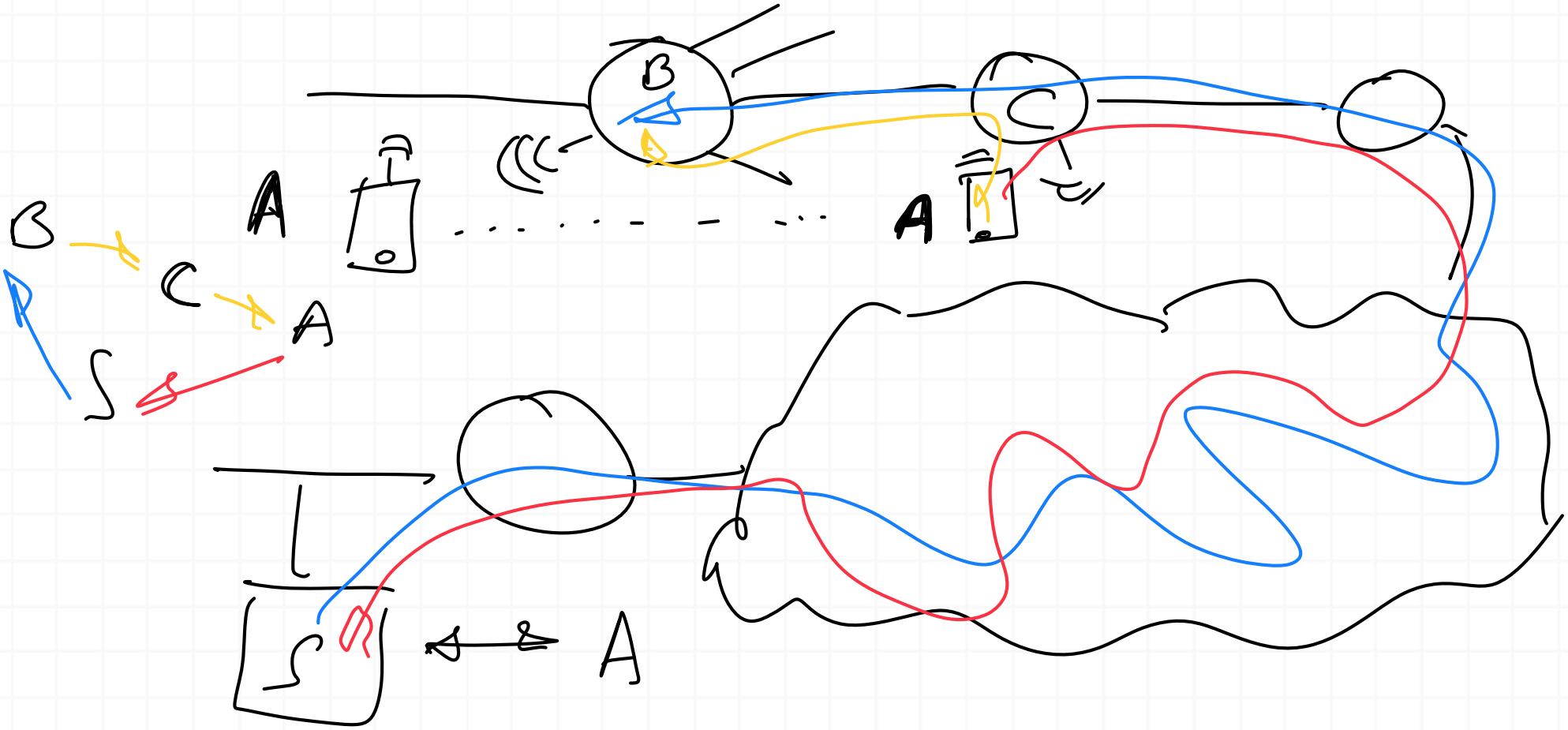
Dr. Ing. J. Ignacio Alvarez-Hamelin

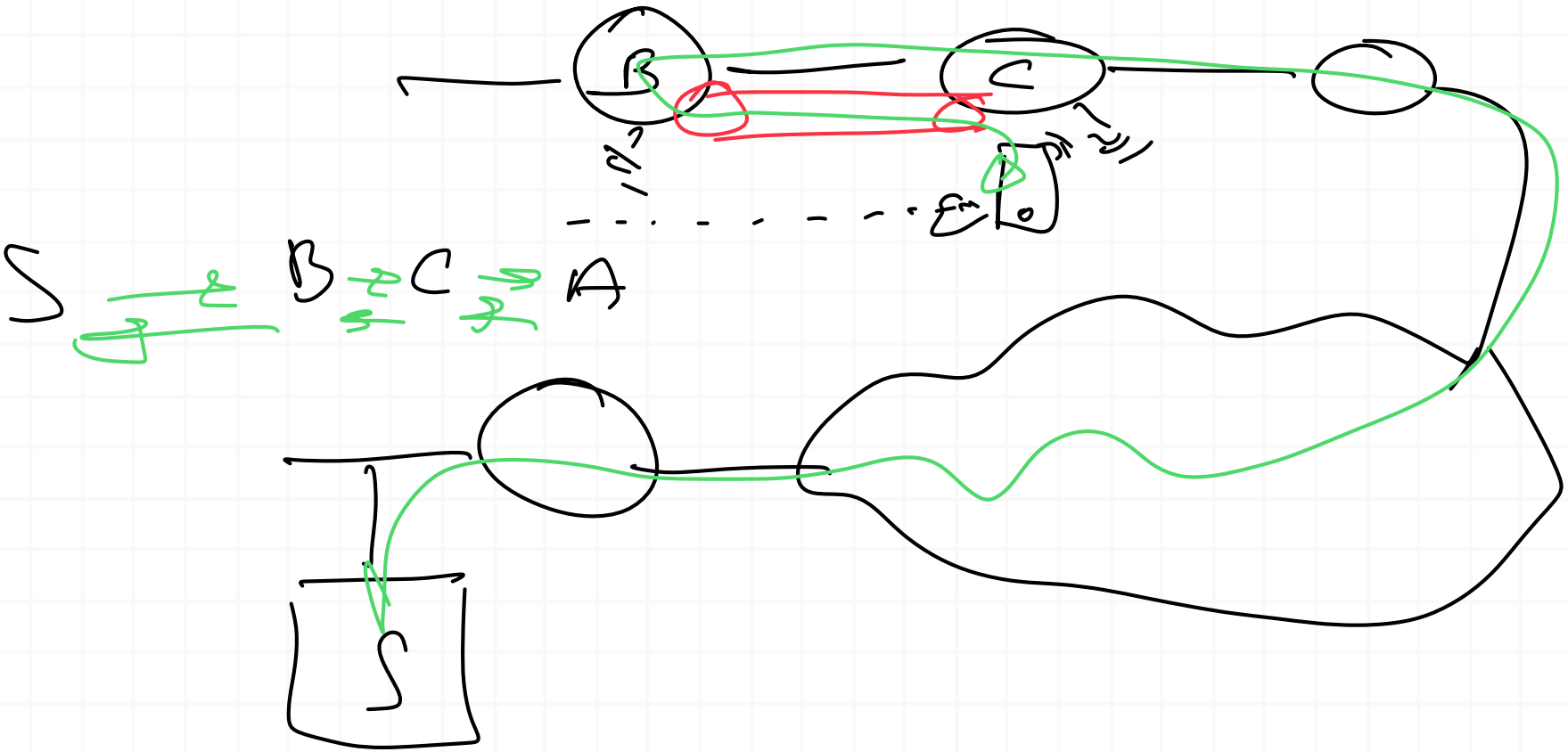
Clase de hoy

- Principios de gerenciamiento de la movilidad
- Mobile IPv4 & IPv6
- Movilidad en Redes Celulares
- Acceso a Internet por intermedio de Redes de Celulares

Principios de gerenciamiento de la movilidad

- ¿Es IP un protocolo adaptado a la movilidad?
- Direccionamiento
- Enrutamiento (Indirecto, Directo, transferencia)

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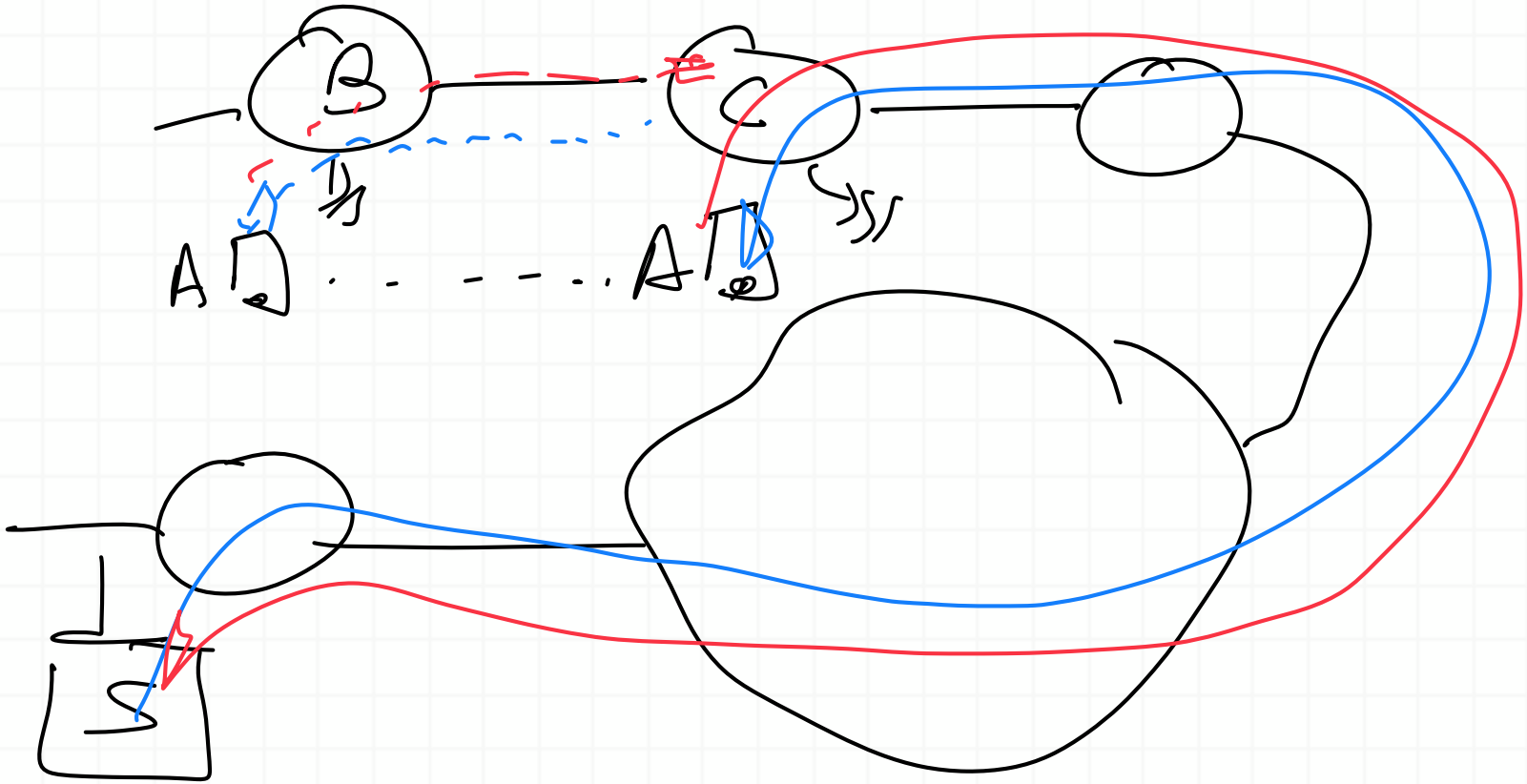
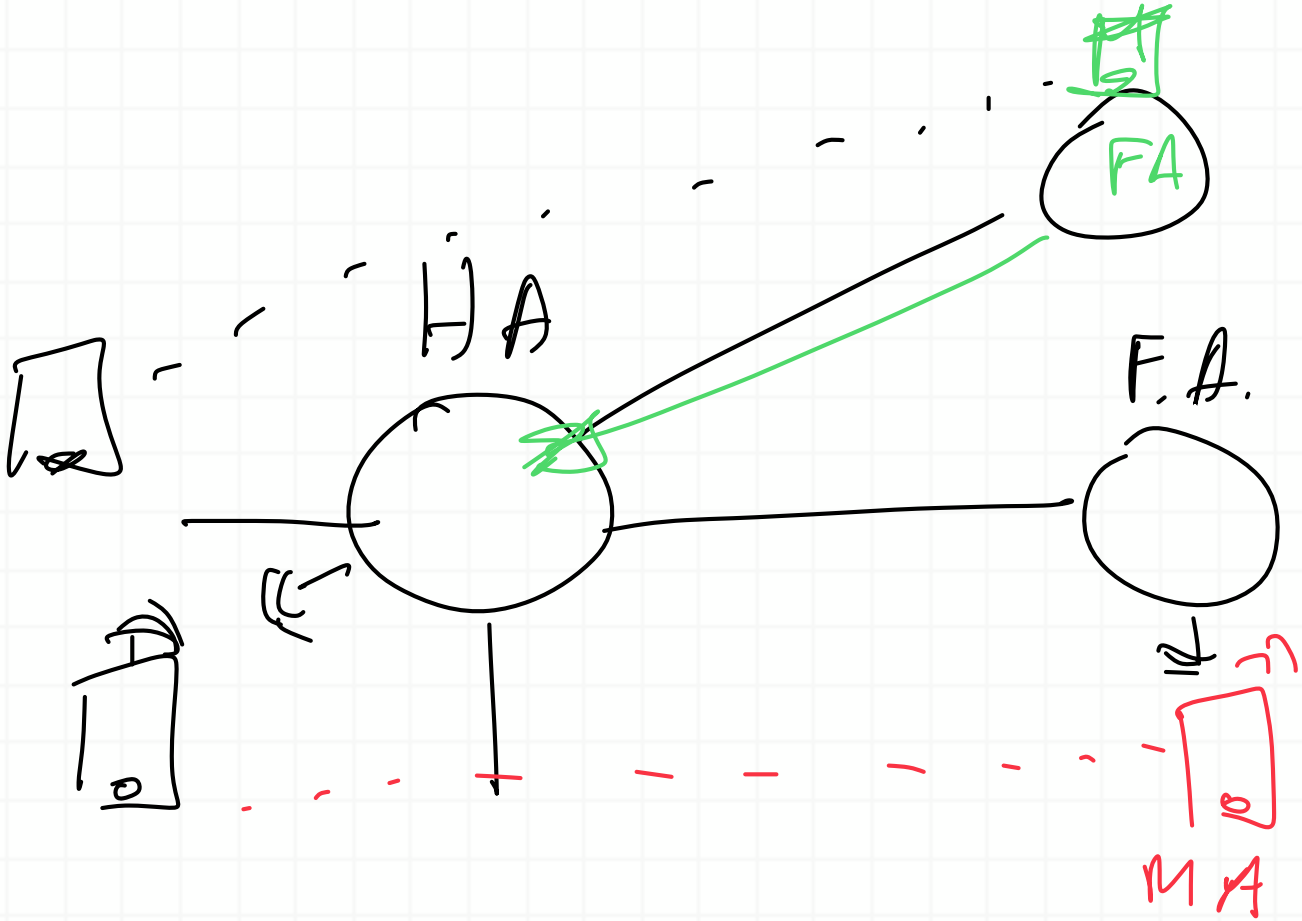


Wi-Max

Mobile IPv4 & IPv6 (RFC 5944, RFC 6275)

- *Mobile Agent*
- *Foreign Agent*
- *Home Agent*



Mobile IPv4 & IPv6 (RFC 5944, RFC 6275)

- *Mobile Agent*
- *Foreign Agent*
- *Home Agent*

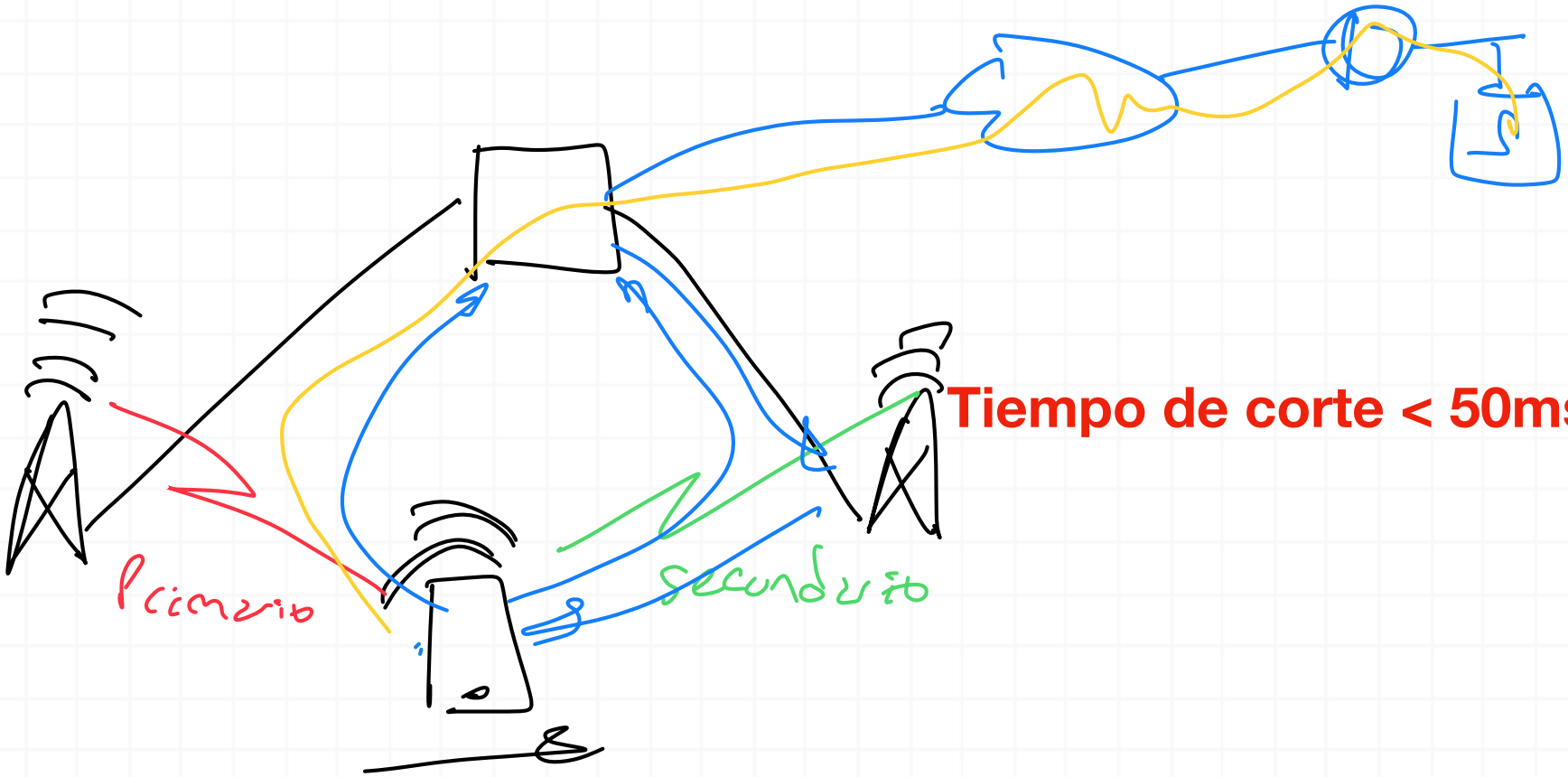
```
                2) Datagram is intercepted   3) Datagram is
                   by home agent and            detunneled and
                   is tunneled to the           delivered to the
                   care-of address.             mobile node.

                      +-----+          +-------+         +------+
                      |home | =======> |foreign| ------> |mobile|
                      |agent|          | agent | <------ | node |
                      +-----+          +-------+         +------+
   1) Datagram to    /|\         /
      mobile node     |        /   4) For datagrams sent by the
      arrives on      |      /        mobile node, standard IP
      home network    |    /          routing delivers each to its
      via standard    |  |_           destination.  In this figure,
      IP routing.   +----+            the foreign agent is the
                    |host|            mobile node's default router.
                    +----+
```

Figure 1: Operation of Mobile IPv4

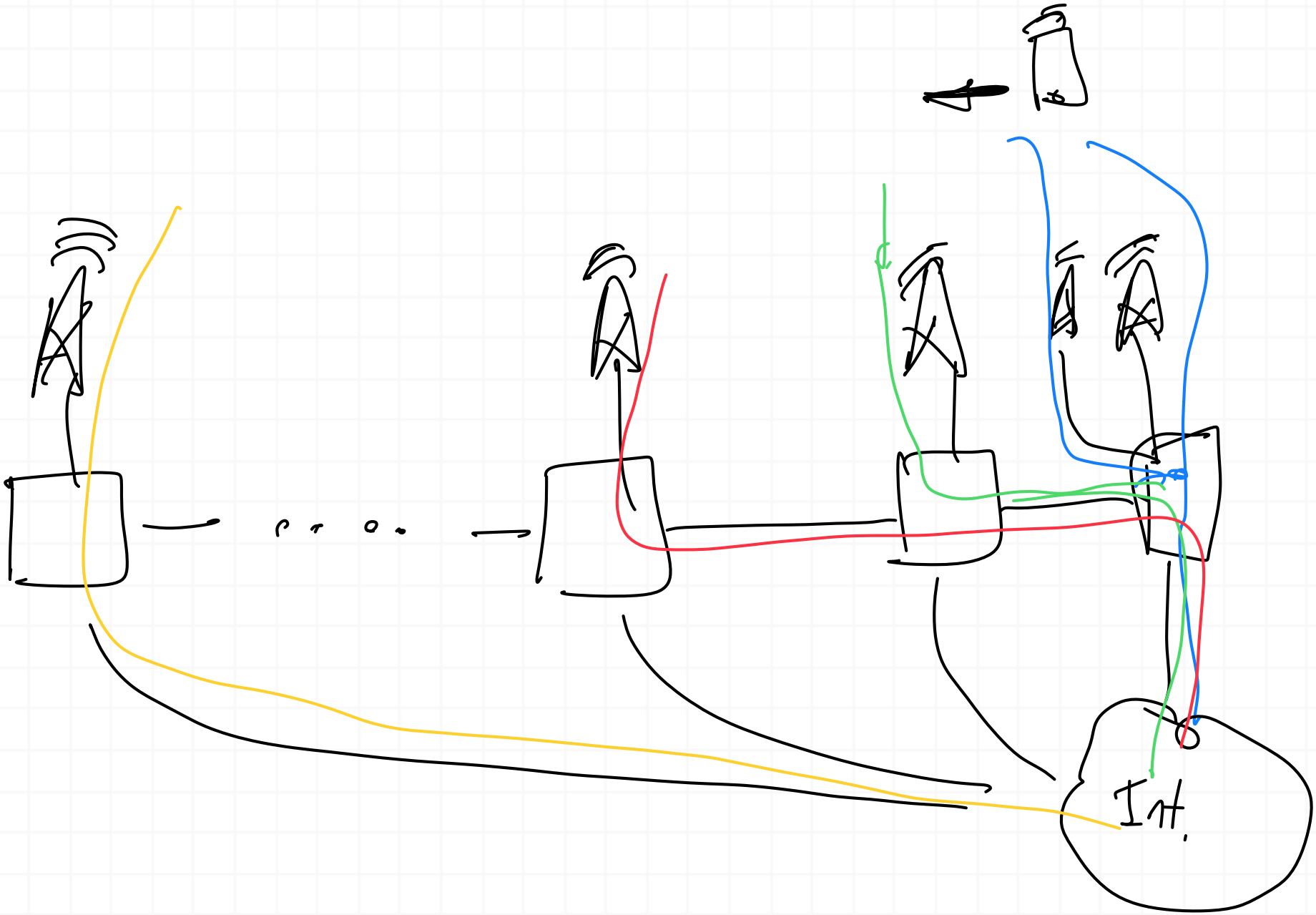
Movilidad en Redes Celulares

- *Handoffs*
- *Rerouting*



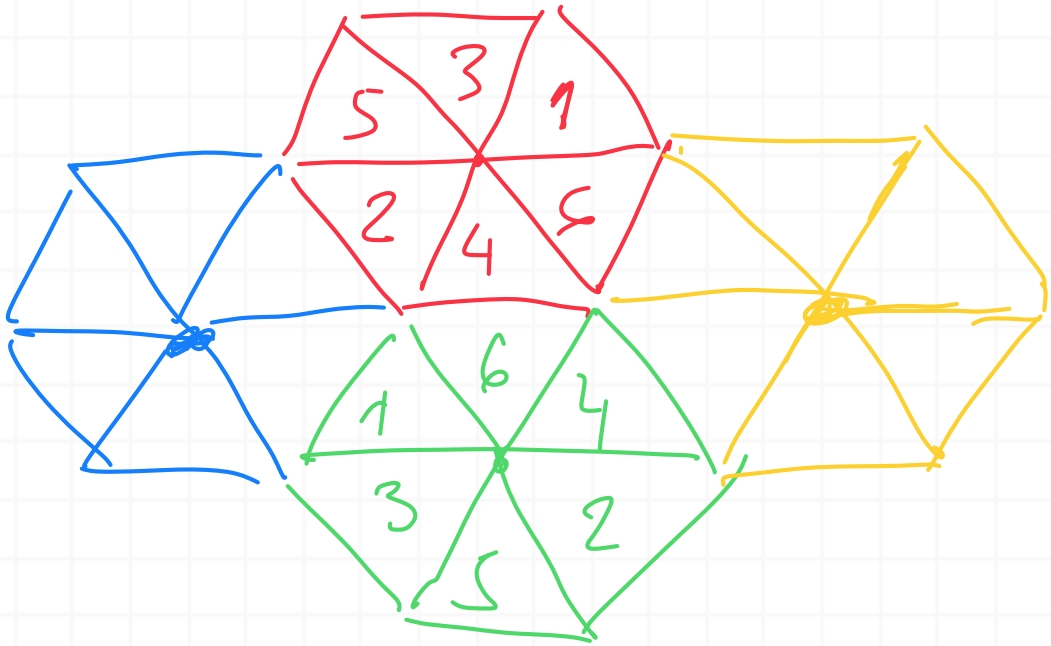
Tiempo de corte < 50ms

3001#12345#


Int.

Acceso a Internet por intermedio de Redes de Celulares

- Desde 1G hasta 5G
- Disposición geográfica
- Tecnologías



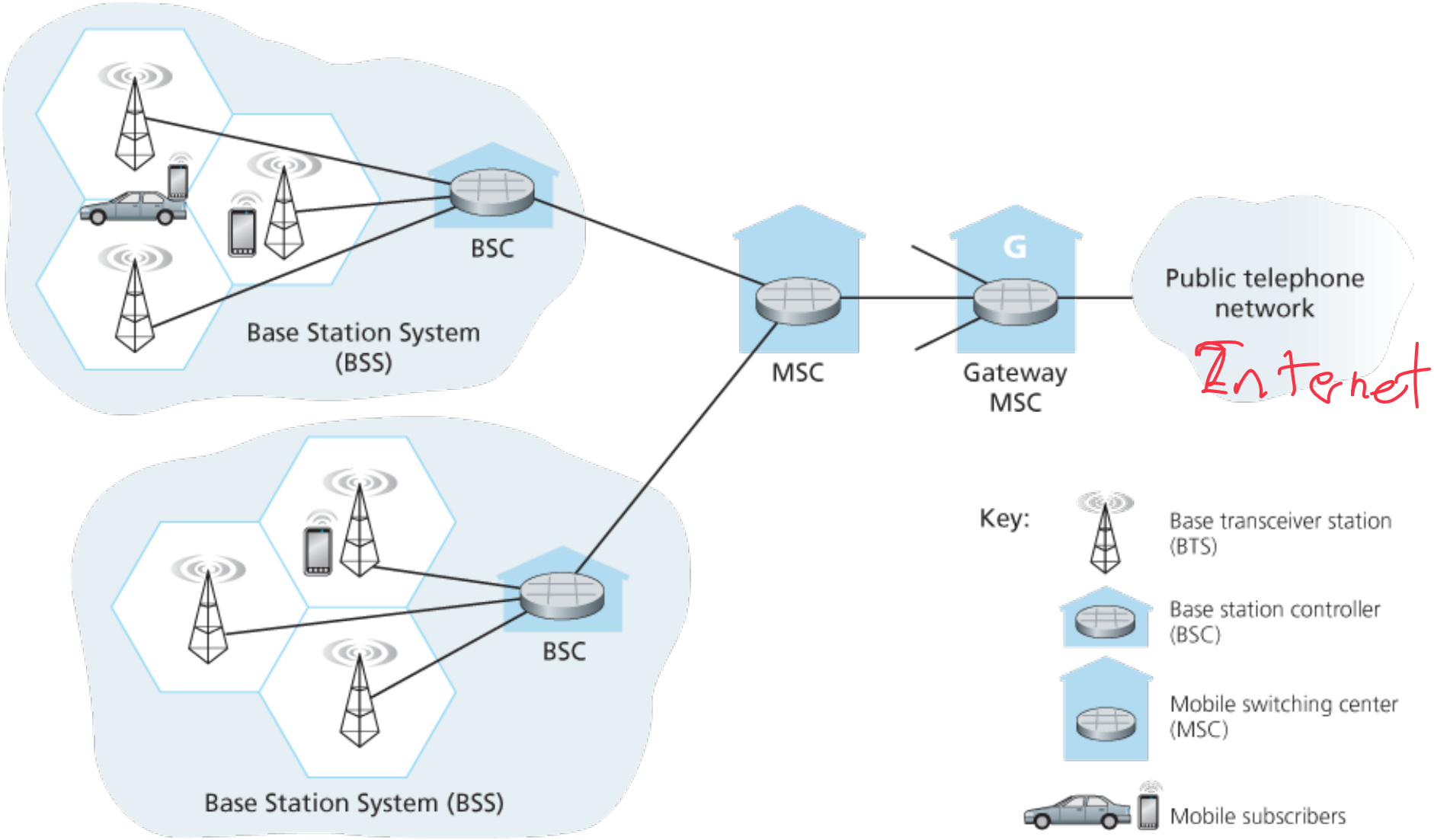


Figure 7.18 Components of the GSM 2G cellular network architecture

A Top-Down Approach with Access. James Kurose and Keith Ross. Publisher: Pearson, Edition: 7th, 2016.

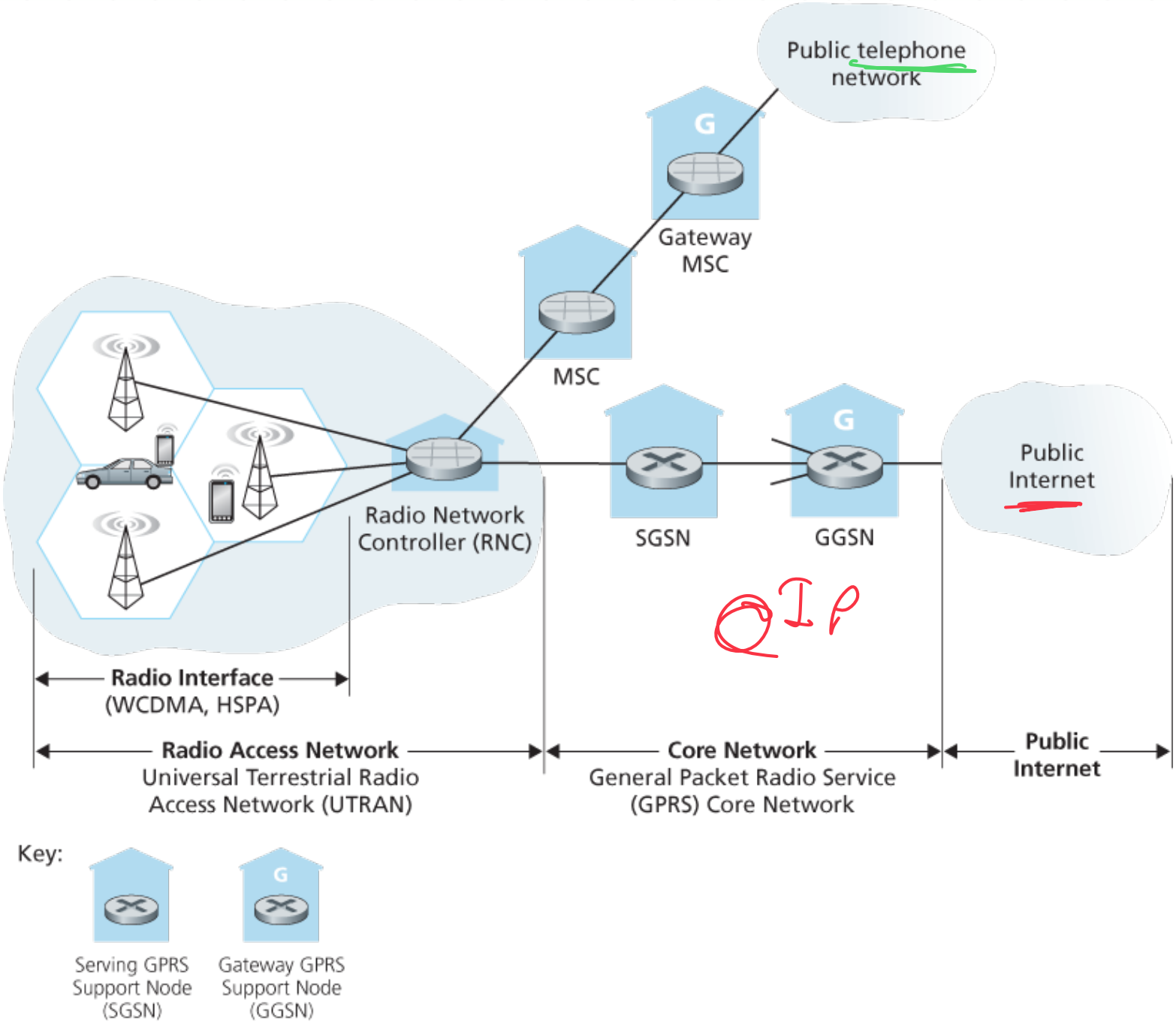


Figure 7.19 3G system architecture

A Top-Down Approach with Access. James Kurose and Keith Ross. Publisher: Pearson, Edition: 7th, 2016.

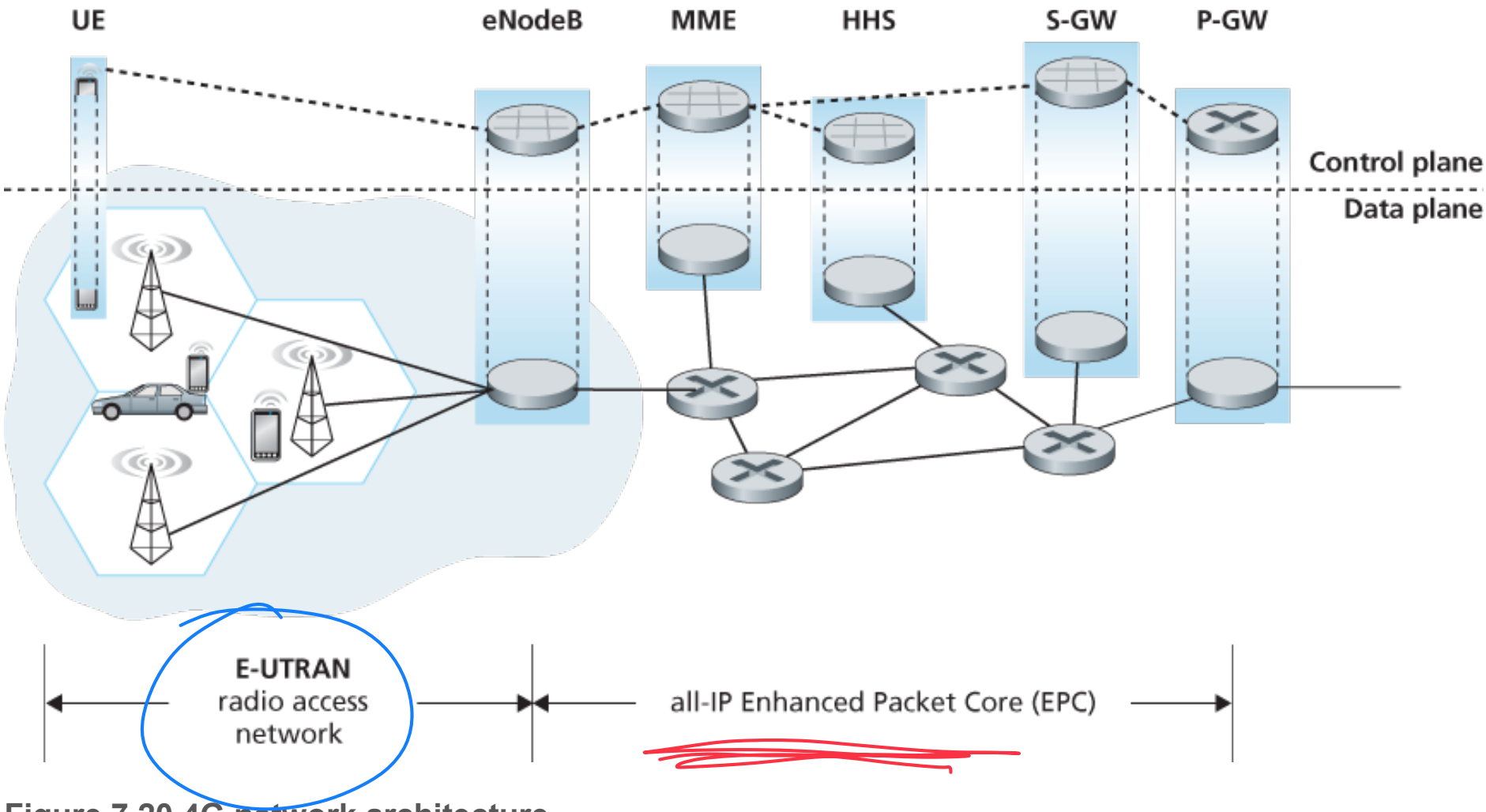


Figure 7.20 4G network architecture

A Top-Down Approach with Access. James Kurose and Keith Ross. Publisher: Pearson, Edition: 7th, 2016.

¿Cómo afecta la movilidad en los protocolos superiores?

UDP : dependerá de la aplicación

TCP : cree que hay algún mecanismo de congestión

Lectura para la próxima clase: Capítulo 8 completo del libro *Computer Networking: A Top-Down Approach. James Kurose and Keith Ross. Publisher: Pearson, Edition: 8th, 2021.*